

WHAT NEEDS TO HAPPEN: FOUR POLICY MOVES THAT MAKE SENSE

For C-V2X to take hold in India, the technology also needs policy support. There are a few straightforward steps that could accelerate deployment and make connected mobility a meaningful part of India's road safety strategy.

1. License-Exempt the Safety Spectrum

The 5.9 GHz band is reserved for road safety worldwide. In China, Europe, Korea, and Canada, V2X devices don't need a license. India should follow suit. Qualcomm argues that treating this safety spectrum as a "gift to society" will be a catalyst for mass adoption.

2. Include V2X in Bharat NCAP

Crash test programs around the world already award points for V2X. Bharat NCAP, launched in 2023, could do the same. Automakers care about safety ratings. Ratings affect sales. Incentives matter.

3. Prioritize Two-Wheelers for Rollout

Install low-cost V2X transmitters on all new motorcycles. It's the easiest way to make India's most vulnerable vehicles more visible. Over time, once adoption stabilizes, we can move to advanced V2X features for real-time collision alerts.

4. Equip Emergency and Heavy Vehicles

Ambulances and police vehicles should be first in line. In India, emergency response times average 33 minutes in rural areas. In Japan, it's 8.5 minutes. With V2X integration, emergency vehicles could signal traffic lights and carve their own green corridors in real-time.

Heavy vehicles, too, are a priority. Trucks account for 20% of road deaths, buses for another 6.5%. C-V2X provides early alerts—critical when dealing with massive, slow-reacting vehicles.

One of the most impactful aspects of V2X is its ability to protect vulnerable road users, especially two-wheelers, cyclists, and pedestrians, by making them more visible to nearby vehicles. On the traffic management front, V2X can optimise signal timing to improve flow, reduce idle times, and lower emissions. And by providing real-time congestion data, it helps drivers make smarter routing decisions while also preventing incident-related bottlenecks in the first place.

Why India is perfect for V2X

India's two-wheeler ecosystem presents a mix of scale and vulnerability. Around 90% of motorcycles sold in the country fall into the entry-level segment, and two-wheelers are involved in nearly half of all road fatalities.

One practical way forward is to equip new motorcycles with basic transmit-only V2X units. These compact devices simply broadcast the vehicle's location, speed, and direction, which is enough to alert nearby cars, trucks, or buses of their presence. It's a low-cost solution, but one with potential for real impact, especially in

urban traffic where visibility and reaction times are crucial.

What works in India's favour is that domestic automakers are no strangers to the tech. Many already know how to integrate C-V2X into vehicles. The industry readiness is there. The next step would be enabling frameworks and policies to move things forward.

Leading this shift is Mahindra, currently the only Indian automaker piloting V2X capabilities in its production vehicles. In collaboration with Qualcomm, Mahindra is testing vehicle-to-vehicle (V2V) features that offer real-time communication between cars, a promising step that underlines both local feasibility and OEM commitment.

Infrastructure is the last piece of the puzzle

While much of the focus around C-V2X tends to sit with vehicles, the infrastructure side of the equation is just as important. To unlock the full potential of the technology, roads and intersections need to be part of the network.

One practical step is to equip urban traffic signals with C-V2X

radios. These systems can send I2V (Infrastructure-to-Vehicle) alerts such as red-light violation warnings, slippery road conditions, or even early notifications about lane closures and construction zones. The goal is to make roads more communicative, not just reactive.

India doesn't have to build a new framework altogether. Countries like the United States have already developed V2I (Vehicle-to-Infrastructure) models that India could adapt to suit its conditions. The key is to ensure consistency, which includes shared standards, uniform interfaces, and interoperable alerts across locations.

Looking ahead

The idea that no road fatality is acceptable—known globally as Vision Zero—is no longer just a European aspiration. It's a mindset that India could adopt in a way that's practical and scalable.

The technology exists, the industry is familiar with it, and the global playbook is clear. We need a push from policymakers, regulators, and public infrastructure planners to move C-V2X to large-scale deployment. ■